

Arithmetic Operations with a Quantum Touch

The discipline of computing is an ever-evolving field that continues to develop and grow as time goes on. In the past we have seen how novel problems breed the need for innovations that can efficiently resolve the challenges that continue to appear with the passage of time and the evolution of our understanding. In our modern age, this can be attributed to quantum computing, as it is both the problem and the solution. As it stands currently, the field of quantum computing is not often taught to the new generations of computer scientists, as it is often relegated as optional material that can be covered if a student seeks it out. As we reach the limits of classical computing, we begin to see why this outdated approach can hinder our quest for progress, as the scope for research is trapped in a decreasing realm of possibility. Quantum computing can provide a fresh input of ideas that propose many solutions for problems that plague us to this day, like optimization, encryption, and space management. For this reason, we here at the UPRM have decided that we wanted to implement and design quantum circuits that realize basic operations, with the goal of proving that quantum mechanics can be applied to everyday operations. To do this, we have used the python library Qiskit to implement a series of arithmetic circuits, each realizing basic mathematical operations (addition, subtraction, multiplication, and division) to carry out a set of example exercises. For the first set, we searched and implemented theoretical arithmetic circuits that were postulated in different papers, which provided a foundation for our work. Then we implemented the same 4 circuits using classical circuits as examples, with the goal of comparing the efficiency and performance metrics between the two. This serves as both a foundational learning exercise, that can impulse us to greater projects, and as proof that quantum computing can be made accessible to many computer science students with the help of programming languages like python.